

$$D \nabla^4 w(x, y, t) + \rho h \frac{\partial^2 w(x, y, t)}{\partial t^2} = q(x, y, t)$$

$$\nabla^4 w = \frac{\partial^4 w}{\partial x^4} + 2 \frac{\partial^4 w}{\partial x^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4}, \quad D = \frac{E h^3}{12 (1 - \nu^2)}$$

w : transverse displacement; q : load per unit area; ρ : density; h : thickness; E : Young's modulus; ν : Poisson's ratio; D : bending stiffness